

not only speeds up your development process but also offers tools to automate testing. Here, we will discover some of the best tools from Django to automate app testing. But before we do that, let's have a quick look at some benefits of testing automation.

Advantages of Testing Automation

Let's consider testing a local library web app. It will have pages for books and authors. These pages have detailed views of the authors and books as objects. Though it is a small web app with a relatively low number of pages, imagine the time it could take to test all the pages and its content.

- Automated testing is faster than the manual testing process.
- Automated testing helps development teams test to a granular level.
- Simulating the end-user experience becomes easier with testing automation.
- Automated testing often becomes the basis for code examples and documentation.
- Automated testing helps with test-driven and behavior-driven tests

Now that we have a grasp on the advantages of choosing testing automation, let's see how it could be done with Django apps.

Types of tests

There are many different types of tests for a web application. They help development teams understand multiple errors and the system's behavior in response to a user request. The following are some crucial tests:

- Unit testing
- Regression testing
- Integration testing
- Functionality testing

Unit Testing

Unit testing concerns measuring the performance of small pieces of code in your web applications. For such testing, short pieces of code like class, module, or even objects, as that of a Django web app, are taken.

With unit testing, the whole process becomes reliable and efficient. As the testing becomes granular, the chances of errors reduce. However, the only problem with unit testing remains the amount of time it may require. Since every small unit of the code is tested, the testing time increases exponentially.